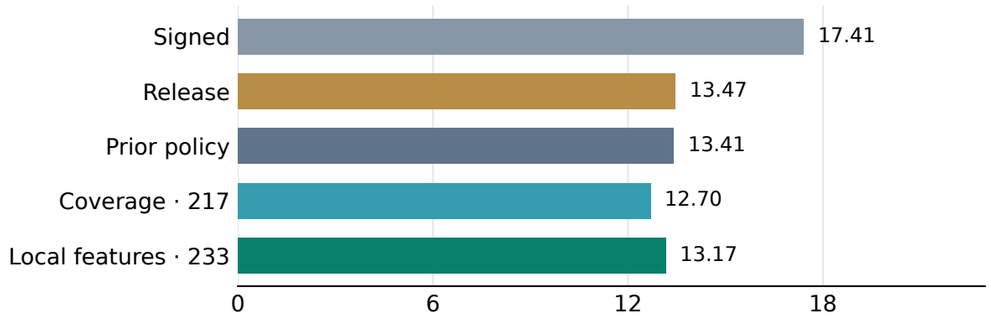
# Unified budget decisions: coverage and carry features

Two matched 300-tree fits · identical 1,889 training circuits · every cap reduction requires learned corroboration

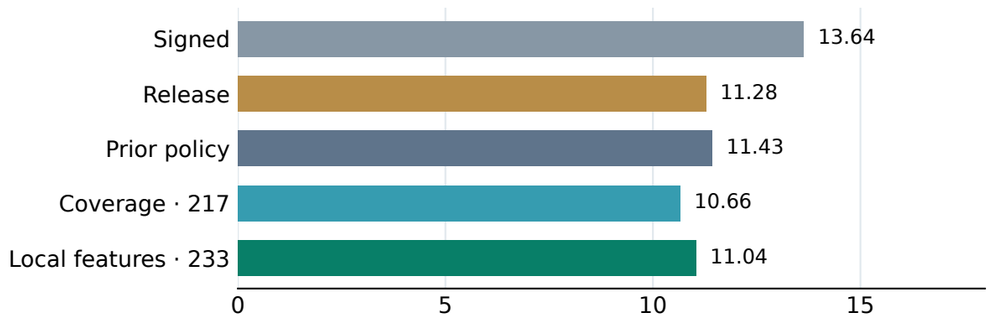
**Development selection: none (development gates failed) · Evaluated: Local features · 233 · Not eligible for promotion**

## General cohort · 32 fixed circuits

Policy MAE · LUT4 ↓




Policy positive-target MAPE · % ↓



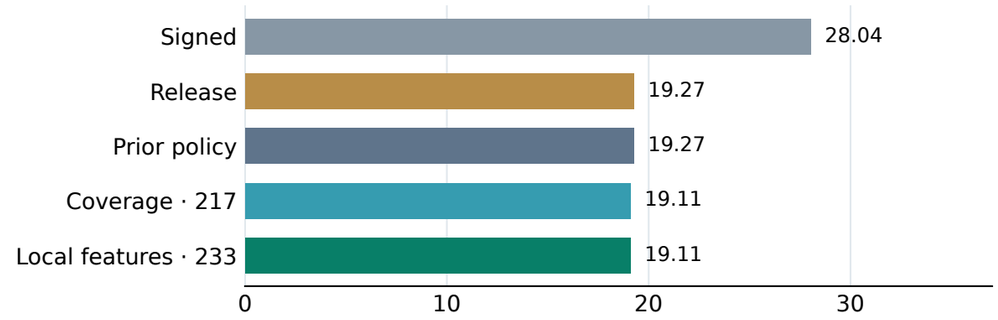

**False accepts / false rejects · each budget assessed separately**

| Cap | Signed | Release | Prior | 217 | 233 |
|---|---|---|---|---|---|
| 32 | 0 / 1 | 0 / 1 | 0 / 1 | 0 / 1 | 0 / 1 |
| 64 | 4 / 0 | 0 / 0 | 2 / 0 | 0 / 0 | 0 / 0 |
| 128 | 6 / 0 | 1 / 5 | 1 / 3 | 1 / 2 | 1 / 3 |

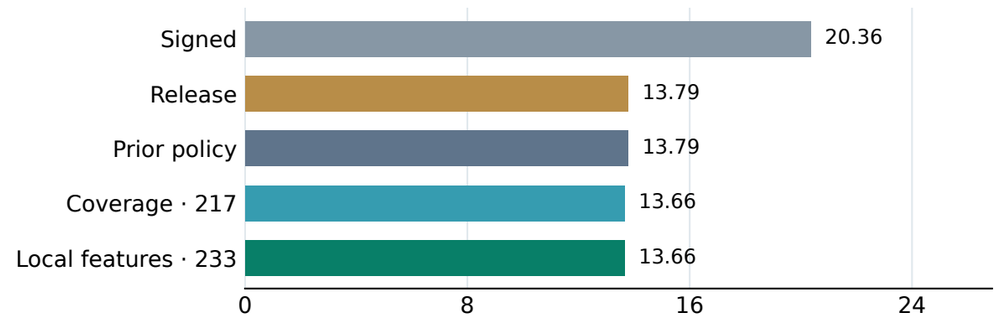
Feasible / infeasible: 32: 1/31 · 64: 8/24 · 128: 20/12

## Conditional cohort · 1 active + 8 shared controls

Policy MAE · LUT4 ↓




Policy positive-target MAPE · % ↓




**False accepts / false rejects · each budget assessed separately**

| Cap | Signed | Release | Prior | 217 | 233 |
|---|---|---|---|---|---|
| 32 | 0 / 0 | 0 / 0 | 0 / 0 | 0 / 0 | 0 / 0 |
| 64 | 2 / 0 | 0 / 0 | 0 / 0 | 0 / 0 | 0 / 0 |
| 128 | 3 / 0 | 1 / 1 | 1 / 1 | 1 / 0 | 1 / 0 |

Feasible / infeasible: 32: 0/9 · 64: 2/7 · 128: 6/3

**Mandatory sharing cases · n=8 · release → evaluated candidate: MAE 30.02 → 30.02 LUT4 · FA/FR at 128: 1/0 → 1/0**

**Mandatory legacy-disagreement cases · n=8 · release → evaluated candidate: MAE 26.02 → 25.33 LUT4 · FA/FR at 64: 1/0 → 1/0**

Fresh activity: 1 action(s) / 1 template(s); legacy-64 vetoes: 1; sharing-128 vetoes: 0. Eight controls shared; no pooling.

Measured target: isolated iCE40 LUT4 count. This is not a CPU speedup, application-transfer or universal hardware-safety result.